

Problem

The current in an RLC circuit is described by

$$\frac{d^2 i}{dt^2} + 10 \frac{di}{dt} + 25i = 0$$

If $i(0) = 7$ A and $di(0)/dt = 0$, find $i(t)$ for $t > 0$.

Step-by-step solution

Step 1 of 3

The current in the RLC circuit described by the second order differential equation is,

$$\frac{d^2 i}{dt^2} + 10 \frac{di}{dt} + 25i = 0 \quad \dots\dots (1)$$

The initial current conditions are, $i(0) = 7$ A and $\frac{di(0)}{dt} = 0$.

Applying Laplace Transform to the equation 1 on both sides

$$\left[s^2 I(s) - si(0) - \frac{di(0)}{dt} \right] + 10 \left[sI(s) - i(0) \right] + 25I(s) = 0$$

Substitute 7 A for $i(0)$ and 0 for $\frac{di(0)}{dt}$.

$$\left[s^2 I(s) - 7s - 0 \right] + 10 \left[sI(s) - 7 \right] + 25I(s) = 0$$

$$(s^2 + 10s + 25)I(s) - 7s - 70 = 0$$

$$(s^2 + 10s + 25)I(s) = 7s + 70$$

$$I(s) = \frac{7(s+10)}{s^2 + 10s + 25}$$

$$I(s) = \frac{7(s+10)}{(s+5)^2}$$

Step 2 of 3

Take partial fractions.

$$I(s) = \frac{A}{s+5} + \frac{B}{(s+5)^2} \dots\dots (2)$$

Find the values of A and B.

$$\frac{7(s-10)}{(s+5)^2} = \frac{A}{s+5} + \frac{B}{(s+5)^2}$$

$$7s + 70 = A(s+5) + B$$

Compare coefficients s.

$$A = 7$$

Compare coefficients of constants.

$$70 = 5A + B$$

$$B = 70 - 5A$$

$$= 70 - 5(7)$$

$$= 35$$

Step 3 of 3

Substitute 7 for A and 35 for B in equation 2.

$$I(s) = \frac{7}{s+5} + \frac{35}{(s+5)^2}$$

Apply Inverse Laplace Transform.

$$L^{-1}[I(s)] = L^{-1}\left[\frac{7}{s+5}\right] + L^{-1}\left[\frac{35}{(s+5)^2}\right]$$

$$i(t) = 7e^{-5t}u(t) + 35te^{-5t}u(t)$$

$$= (7 + 35t)e^{-5t}u(t) \text{ A}$$

$$= (7 + 35t)e^{-5t} \text{ A} \quad \text{for } t > 0$$

Thus, the $i(t)$ for $t > 0$ is $\boxed{(7 + 35t)e^{-5t} \text{ A}}$.